

The Graphics Card Industry

The 1991 to 2000 decade saw the establishment of the PC as a viable entertainment platform - games as well as movies. While consoles from Atari, Nintendo, and Sega were popular in the first half of the decade, with PCs becoming more affordable and the number of homes with PCs increasing, towards the end of the decade PCs had become a force to reckon with.

Graphics cards that existed at that time were primarily for 2D graphics, since 2D was being used by high-end workstations. Video games used 3D graphics, and this created a different market segment. Factors that influenced the graphics card industry included the release of PC-only games like ~Quake~, the introduction of the AGP interface, and the introduction of DirectX as a major graphics API in a market with many proprietary graphics API like 3dfx's Glide.

3dfx was the first graphics card manufacturer that released a card specifically for gaming, in the form of the Voodoo, in 1996. Being unable to deal with 2D load, systems required a separate graphics card in addition to the Voodoo, but the card proved popular nevertheless. 3dfx saw considerable success with every version of the Voodoo series. The last in the series was the Voodoo 5.

NVIDIA, whose first chip, the NV1, used a different graphics rendering method from that supported by DirectX, had to create an entirely new design that was released as the Riva 128. NVIDIA followed this up with the Riva TNT, and then the Riva TNT2... then the GeForce 256 and GeForce 2 families. The Riva TNT2 proved too good a match for the 3dfx Voodoo 2, and helped NVIDIA become market leader. In 2000, NVIDIA took over 3dfx.

Through these years, ATI released the Mach and Rage series of graphics cards which helped it to maintain its position in the market. This situation changed with the Radeon series, when it began challenging NVIDIA.

Besides these, there were others like Cirrus Logic, S3, Matrox, SiS, Intel, and 3D labs. S3 was acquired by Via and their know-how was used to create onboard graphics for Via chipsets; Intel and SiS gave up their pursuits in the field of discrete cards and focused on offering onboard solutions. Matrox focused on 2D graphics.